Intro to Mathematical Modeling: Shapes of Graphs & Variables

1. Thinking About Shapes of Graphs: Partnership Life Proportion

Problem Statement: Suppose that a person meets their future partner at age 18, that the partner dies when the person is 70, and that the person then goes on to live to age 100. If x represents the person's age, draw the graph showing the proportion $P(x)$ of their life that they have been with their partner. Then write down a formula for $P(x)$, showing all your work.

Mathematical Formulation & Formula Derivation:

- **Phase 1 ($x < 18$):** The person has not yet met their partner. The time spent together is 0.

$$P(x) = 0$$

- **Phase 2 ($18 \leq x \leq 70$):** The person met their partner at age 18. At age x, the total number of years spent together is $x - 18$. The person's total lifetime lived to date is x. Therefore, the proportion is:

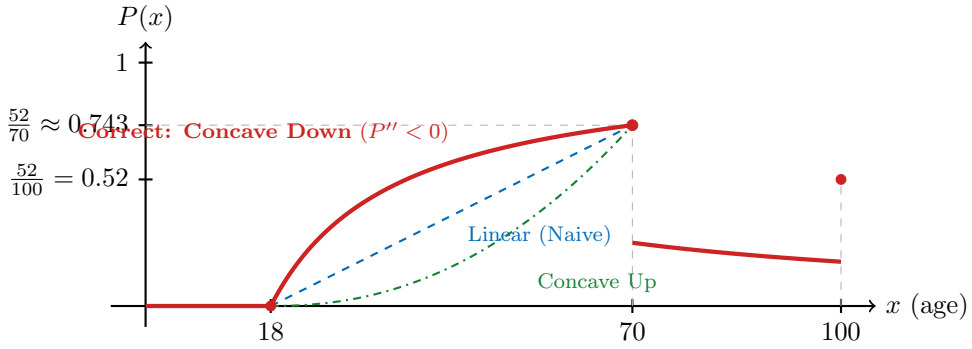
$$P(x) = \frac{x-18}{x} = 1 - \frac{18}{x}$$

- **Phase 3 ($70 < x \leq 100$):** The partner died at age 70. The total time spent together is frozen at $70 - 18 = 52$ years. The person continues to age to x. Therefore, the proportion of their life spent together is:

$$P(x) = \frac{52}{x}$$

$$P(x) = \begin{cases} 0 & \text{if } x < 18 \\ \dfrac{x-18}{x} = 1 - \dfrac{18}{x} & \text{if } 18 \leq x \leq 70 \\ \dfrac{52}{x} & \text{if } 70 < x \leq 100 \end{cases}$$

Graph Analysis & Justification of Concavity:



- **Why is $P(x)$ concave down on $[18, 70]$?** Taking derivatives of $P(x) = 1 - 18x^{-1}$:

$$P'(x) = \frac{18}{x^2} > 0 \quad \text{(strictly increasing)}, \qquad P''(x) = -\frac{36}{x^3} < 0 \quad \text{(strictly concave down)}.$$

Intuitive Reasoning: Immediately after meeting at age 18, each additional year lived adds a massive fraction to the proportion. By age 36 (just 18 years later), the person has spent $18/36 = 50\%$ of their life with their partner. But in later years, adding one year increases the fraction only marginally (at age $69 \to 70$, the change is under 0.4%). Thus, the rate of increase slows down, producing a concave down curve.

- **Why does $P(x)$ decrease and curve concave up on $(70, 100]$?** $P'(x) = -52x^{-2} < 0$ and $P''(x) = 104x^{-3} > 0$. The proportion dilutes toward zero as the total lifespan x grows while the shared years remain fixed at 52.

2. Qualitative Modeling: Contagious Disease Epidemic

Problem 1: Consider a situation in which a contagious disease is spreading through a city of $1,000,000$ people. Assume also that:

(a) no one (or almost no one) dies from the disease,

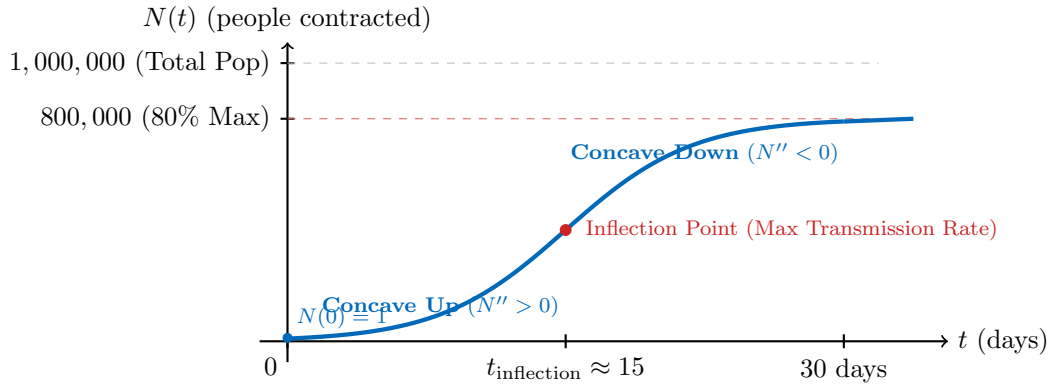
(b) once one recovers from it, one doesn't get the disease again (permanent immunity),

(c) the epidemic runs its course in a one month (30 day) period,

(d) 20% of the population never gets the disease, and

(e) the epidemic begins with one person ($N(0) = 1$).

Draw a possible graph that gives the number of people who have *contracted* the disease during the epidemic as a function of time. Label axes carefully (including scales) and justify the shape of the graph.



Detailed Justification of the S-Shape (Logistic Growth):

- **Initial Condition & Cumulative Measure:** $N(t)$ measures total people who have *contracted* the disease. Since no one loses contracted status (immunity means they don't get it again, and no one dies), $N(t)$ is strictly non-decreasing ($N'(t) \geq 0$). At $t = 0$, $N(0) = 1 \approx 0$.
- **Early Exponential Phase (Concave Up, $N'' > 0$):** Initially, almost the entire city is susceptible ($999,999$ people). As the number of infected individuals grows, there are more transmitters interacting with a virtually unlimited pool of susceptible hosts. The number of new daily infections accelerates, so the slope steepens ($N'' > 0$).
- **Inflection Point (Peak Spread Rate):** Around the middle of the month (approximately day 15), the spread rate dN/dt reaches its maximum.
- **Late Saturation Phase (Concave Down, $N'' < 0$):** As the number of contracted individuals approaches the threshold of $800,000$ (80% of $1,000,000$), infected people encounter fewer and fewer susceptible people. The spread decelerates, causing the curve to become concave down ($N'' < 0$) and level off asymptotically toward $800,000$.
- **Endpoint ($t = 30$):** At $t = 30$ days, the epidemic has run its course, meaning new infections drop to 0 and the graph flattens horizontally at exactly $N = 800,000$.

3. Playing with Variables: Gas Mileage & Travel Optimization

Problem 2: Suppose you have a car that gets 24 miles per gallon and you are going to buy 10 gallons of gas. A gas station near you sells gas for \$2.499/gal, but one 4 miles down the road sells gas for \$2.459/gal. Assuming gas is the only cost:

(a) Would you save money by driving the extra 4 miles?

(b) What if the gas station were only 1 mile away?

(c) What is the least distance at which you would *not* save any money?

General Algebraic Cost Model: Let D be the one-way distance to the further gas station (miles).

- Near station cost: $C_{\text{near}} = 10 \text{ gal} \times \$2.499/\text{gal} = \mathbf{\$24.99}$.

- Round-trip distance to further station: $2D$ miles.
- Gas consumed on round-trip: $\dfrac{2D \text{ miles}}{24 \text{ mpg}} = \dfrac{D}{12}$ gallons.
- Total gas purchased: To end up with 10 net gallons in the tank, you must purchase $10 + \dfrac{D}{12}$ gallons at \$2.459/gal:

$$C_{\text{far}}(D) = \left(10 + \frac{D}{12}\right) \times 2.459 = 24.59 + \left(\frac{2.459}{12}\right) D \approx 24.59 + 0.204917D$$

Solutions to Subquestions:

(a) **Distance $D = 4$ miles:**

$$C_{\text{far}}(4) = 24.59 + \frac{4}{12} \times 2.459 = 24.59 + \frac{2.459}{3} \approx 24.59 + 0.82 = \$\mathbf{25.41}$$

Since \$25.41 > \$24.99, you would **lose** \$0.42. **No, you do not save money.**

(b) **Distance $D = 1$ mile:**

$$C_{\text{far}}(1) = 24.59 + \frac{1}{12} \times 2.459 \approx 24.59 + 0.205 = \$\mathbf{24.795} \approx \$\mathbf{24.80}$$

Since \$24.80 < \$24.99, you **save** approximately \$24.99 − \$24.795 = \$**0.195** (≈ 20 cents). **Yes, you save money.**

(c) **Break-Even Distance D^*:** Set $C_{\text{far}}(D^*) = C_{\text{near}}$:

$$24.59 + 0.204917D^* = 24.99 \implies 0.204917D^* = 0.40 \implies D^* = \frac{0.40}{0.204917} = \frac{0.40 \times 12}{2.459} \approx \mathbf{1.952} \text{ miles}$$

Conclusion: At any distance $D \geq \mathbf{1.95}$ miles, you will **not** save any money.